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B.sc Information Science CSC 392 : Software Designer & Software Architect

A software designer in software engineering is a professional who plans and designs how the software system will be structured and how it has to function before it is developed and implemented. This means that the software designer works cooperatively with both software architects and programmers in the structuring and running of the system. The software designer develops an actual blueprint that guides the programmer during system development. One major function of a software developer or designer is to analyze user requirements while also ensuring that the system is usable to the user. By analyzing user requirements during system development it is trying to understand what the software has to initially do, its major tasks and functions through analyzing business and end user requirements. For example when developing an information management system for a school, the teachers will expect to upload marks, tests scores and syllabuses in the system while students will need to upload their assignments and view their results, these are user requirements which a system designer has to consider. Upon understanding the user requirements, system designer then converts user needs into specification that system developers are to follow. The specifications are the scope or the modules that are to be included into the system.

Additionally, systems designers have to break down the system modules, define algorithms into smaller components explaining how each has to function and solve problems. On top of that he also has to design user interfaces because for the system to be user friendly it has to have means of interacting with the user like dropdowns and menus while also continuously reviewing the system to see area of improvements during development. Another task is to work closely with developers to clarify the design details of the system so they could understand its structure and functionality. Designers have to design UML diagrams, flow charts and data flow diagrams to explain the logic behind the system. In short the software designer decides how the system will be organized, how different components of the system will interact, how the data will flow within the system and how the user will interact with the system. By system organization it means how the different components of the system cooperate so they work properly. For example a university management system may be organized into lecturer, student, assessment and registry department. All these platforms perform a specific unique task but due to system organization they together form one complete

system.

Software architect The System Architect plays a strategic and high-level role in the development of the Student Hub System. The architect is responsible for defining the overall technical vision and structure of the system before detailed design and coding begin. While other team members focus on specific components, the System Architect concentrates on the entire system environment and ensures that all parts function together as a unified and efficient solution.

One of the main responsibilities of the System Architect is to design the overall architecture of the Student Hub System. This includes deciding how the major components such as the user interface, application logic, and database will interact. For example, the architect may choose a client–server architecture where students access the system through a web browser, the server processes their requests, and the database stores assignment and test information. This decision affects how the entire system will operate and communicate internally. Another key responsibility of the System Architect is selecting the appropriate technology stack. The architect determines which programming languages, frameworks, databases, and hosting platforms will be used. These decisions are made based on performance, scalability, security, and long-term sustainability. For instance, the architect ensures that the selected database can handle many student records efficiently and that the system can support multiple users simultaneously without performance issues.

The System Architect also focuses heavily on scalability and system performance. The Student Hub System may initially serve a small group of students, but it should be capable of expanding to serve the entire university in the future. The architect plans the structure in such a way that new features, such as grade tracking or integration with other academic systems, can be added without redesigning the whole system. Security planning is another critical function of the System Architect. Since the system manages sensitive academic data, the architect designs secure authentication mechanisms, defines access control levels, and ensures that data transmission and storage are protected. These decisions help prevent unauthorized access and protect student information.

It is important to distinguish the System Architect from the Software Designer. The System Architect works at a higher, strategic level and defines the overall system framework and technical standards. In contrast, the Software Designer focuses on the detailed design of individual components, such as designing the layout of the dashboard, defining database fields, or creating system diagrams like ER diagrams and use case diagrams. While the designer concentrates on how specific parts of the system will look and function, the architect ensures that the entire system structure is technically sound and future-ready.

In conclusion, the System Architect is responsible for shaping the overall technical foundation of the Student Hub System. Through architectural planning, technology selection, scalability design, and security structuring, the architect ensures that the system is robust, efficient, and capable of meeting both present and future needs. This high-level responsibility clearly distinguishes the architect's role from that of the Software Designer.

References

Sommerville, I., (2011). Software Engineering 9th edition. ISBN-10,137035152, p.18.